ABSTRACTS AND PAPERS

Academicians, industry professionals, and consultants are invited to present their work at NCRAC 2026. Authors are required to submit an abstract of no more than 250 words via email to ncrac2026@iittp.ac.in. Upon acceptance of the abstract, contributors will be invited to submit a full-length paper, with detailed formatting and submission guidelines to be communicated subsequently. All submissions will be subject to a rigorous peer-review process conducted by a panel of experts from both academia and industry. Dedicated presentation sessions will be organized, including sessions highlighting industry-focused contributions such as field experience, case studies, and emerging trends in the HVAC&R sector.

TOPICS

The conference will cover all significant areas of Refrigeration and Air conditioning, including:

- Energy efficiency & cold/heat recovery
- Novel/alternate ecofriendly RAC systems
- Natural/new refrigerants & mixtures
- Refrigerated cold chain, transport & cold stores
- Sorption heating and cooling systems
- District heating and cooling
- Desiccant and evaporative cooling
- Polygeneration
- Heat pump
- Indoor air quality and Industrial ventilation,
- HVAC systems for Healthcare

IMPORTANT DATES

Short abstracts (250 words)	: 28th Feb 2026
Abstract acceptance	: 15th March 2026
Full paper submission	: 30th June 2026
Revised paper submission	: 30th September 2026
Conference dates	: 13-16th December 2026

REGISTRATION FEE

The registration fee per delegate is ₹12,000, while the fee for an industry observer is ₹5,000. For delegates from academic institutions and research laboratories the registration fee is ₹7,500. The registration fee of ₹3,500. The registration fee for industry participants is ₹15,000. For overseas delegates, the registration fee is USD 400, and for overseas student authors, it is USD 250. Detailed information regarding the registration will be communicated soon.

SPONSORSHIP AND ADVERTISEMENT

In NCRAC 2026, sponsors will have the opportunity to conduct product presentations, exhibit their products, and display posters and catalogues, as well as distribute promotional brochures. The printed conference proceedings will include plenary, keynote, and invited

CALL FOR PAPERS

9th National & 3rd International Conference on

"Refrigeration & Air Conditioning"

NCRAC 2026

13-16th December 2026

Theme: Smart & Sustainable HVAC Systems for Viksit Bharat 2047

organized by

भारतीय प्रौद्योगिकी संस्थान तिरुपति

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OBJECTIVES

The aspirations and living standards of humankind continue to rise alongside rapid technological progress, while challenges such as environmental degradation, global warming, and climate change demand urgent attention. Meeting human comfort and industrial needs in an environmentally sustainable manner places Refrigeration and Air Conditioning (RAC) at the forefront of technological innovation. Beyond human comfort, RAC systems are essential to sectors including food preservation, healthcare, space exploration, and digital infrastructure, and currently account for nearly 20% of global electricity consumption. Although international efforts such as the Montreal Protocol have successfully addressed ozone depletion, the goals of the Kyoto Protocol and Paris Agreement—particularly reducing greenhouse gas emissions and limiting global temperature rise to within 2°C—remain ongoing challenges. Addressing these issues requires the development of eco-friendly refrigerants, innovative system architectures, integration with renewable energy, and life-cycle–based sustainability approaches. This conference aims to bring together researchers, academicians, industry professionals, and policymakers to discuss these challenges, showcase emerging technologies, and chart a path toward a low-carbon, energy-efficient, and sustainable future for refrigeration and air conditioning.

ABOUT NCRAC

ABOUT IIT TIRUPATI

IIT Tirupati, established in 2015, began functioning in the academic year 2015–16 with the support of its mentoring institute, IIT Madras. By 2018, its academic programs had expanded to nine departments spanning engineering, sciences, and humanities, offering B.Tech, M.Tech, M.Sc, M.S.(R), and PhD programmes. Its curricula emphasize a strong foundation in theory complemented by practice-oriented laboratories, fostering innovation, creativity, teamwork, communication skills, ethics, and societal engagement. The permanent campus, spread over 548 acres, is under development, with several buildings operational since 2018. IIT Tirupati is also developing an 18-acre research park on its campus, set to be one of the largest in India upon completion. This facility aims to foster innovation, entrepreneurship, and industry-academia collaborations in areas like smart infrastructure, energy, and AI. Located in Tirupati, the institute benefits from an airport and excellent connectivity to Chennai and Bengaluru.

ABOUT MECHANICAL ENGINEERING DEPARTMENT

The Department of Mechanical Engineering at IIT Tirupati, established in 2015, is a dynamic and rapidly growing academic unit offering

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